

3DModel

A Schema.org Type

This term is in the "new" area - implementation feedback and adoption from applications and websites can help improve our definitions.

Thing > **CreativeWork** > **MediaObject** > **3DModel**

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A 3D model represents some kind of 3D content, which may have **encodings** in one or more **MediaObjects**. Many 3D formats are available (e.g. see [Wikipedia](#)); specific encoding formats can be represented using the **encodingFormat** property applied to the relevant **MediaObject**. For the case of a single file published after Zip compression, the convention of appending '+zip' to the **encodingFormat** can be used. Geospatial, AR/VR, artistic/animation, gaming, engineering and scientific content can all be represented using **3DModel**.

Property	Expected Type	Description
Properties from 3DModel		
isResizable	Boolean	Whether the 3DModel allows resizing. For example, room layout applications often do not allow 3DModel elements to be resized to reflect reality.
Properties from MediaObject		
associatedArticle	NewsArticle	A NewsArticle associated with the Media Object.

Property	Expected Type	Description
bitrate	Text	The bitrate of the media object.
contentSize	Text	File size in (mega/kilo)bytes.
contentUrl	URL	Actual bytes of the media object, for example the image file or video
duration	Duration	The duration of the item (movie, audio recording, event, etc.) in ISO 8601 duration format .
embedUrl	URL	A URL pointing to a player for a specific video. In general, this is the ir in the src element of an embed tag and should not be the same as the ir of the loc tag.
encodesCreativeWork	CreativeWork	The CreativeWork encoded by this media object. Inverse property: encoding
encodingFormat	Text or URL	Media type typically expressed using a MIME format (see IANA site a reference), e.g. application/zip for a SoftwareApplication binary, audio for .mp3 etc.
		In cases where a CreativeWork has several media type representations, encoding can be used to indicate each MediaObject alongside pair encodingFormat information.
		Unregistered or niche encoding and file formats can be indicated inside the most appropriate URL, e.g. defining Web page or a Wikipedia/Wikimedia entry. Supersedes fileFormat .
endTime	DateTime or Time	The endTime of something. For a reserved event or service (e.g. FoodEstablishmentReservation), the time that it is expected to end. For actions that span a period of time, when the action was performed. I

Property	Expected Type	Description
		wrote a book from January to <i>December</i>. For media, including audio and video, it's the time offset of the end of a clip within a larger file. Note that Event uses startDate/endDate instead of startTime/endTime when describing dates with times. This situation may be clarified in future revisions.
height	Distance or QuantitativeValue	The height of the item.
ineligibleRegion	GeoShape or Place or Text	The ISO 3166-1 (ISO 3166-1 alpha-2) or ISO 3166-2 code, the place, or GeoShape for the geo-political region(s) for which the offer or delivery specification is not valid, e.g. a region where the transaction is not allowed. See also eligibleRegion.
interpretedAsClaim	Claim	Used to indicate a specific claim contained, implied, translated or reworded the content of a MediaObject or other CreativeWork . The interpretation can be indicated using claimInterpreter .
playerType	Text	Player type required—for example, Flash or Silverlight.
productionCompany	Organization	The production company or studio responsible for the item, e.g. series, game, episode etc.
regionsAllowed	Place	The regions where the media is allowed. If not specified, then it's assumed to be allowed everywhere. Specify the countries in ISO 3166 format .
requiresSubscription	Boolean or MediaSubscription	Indicates if use of the media require a subscription (either paid or free). Allowed values are true or false (note that an earlier version had 'yes' and 'no').

Property	Expected Type	Description
sha256	Text	The SHA-2 SHA256 hash of the content of the item. For example, a zero input has value <code>'e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7359b3aa997b61122ff071482bb80508b4681ba7412452e771563bd9962ecb1130333'</code>
startTime	DateTime or Time	The startTime of something. For a reserved event or service (e.g. FoodEstablishmentReservation), the time that it is expected to start. For actions that span a period of time, when the action was performed. I wrote a book from <i>January</i> to December. For media, including audio and video, it's the time offset of the start of a clip within a larger file. Note that Event uses startDate/endDate instead of startTime/endTime when describing dates with times. This situation may be clarified in future revisions.
uploadDate	Date or DateTime	Date (including time if available) when this media object was uploaded to the site.
width	Distance or QuantitativeValue	The width of the item.
Properties from CreativeWork		
about	Thing	The subject matter of the content. Inverse property: subjectOf
abstract	Text	An abstract is a short description that summarizes a CreativeWork .
accessMode	Text	The human sensory perceptual system or cognitive faculty through which a person may process or perceive information. Values should be drawn from the approved vocabulary .

Property	Expected Type	Description
accessModeSufficient	ItemList	A list of single or combined accessModes that are sufficient to understand the intellectual content of a resource. Values should be drawn from the approved vocabulary .
accessibilityAPI	Text	Indicates that the resource is compatible with the referenced accessibility API. Values should be drawn from the approved vocabulary .
accessibilityControl	Text	Identifies input methods that are sufficient to fully control the described resource. Values should be drawn from the approved vocabulary .
accessibilityFeature	Text	Content features of the resource, such as accessible media, alternative text, supported enhancements for accessibility. Values should be drawn from the approved vocabulary .
accessibilityHazard	Text	A characteristic of the described resource that is physiologically dangerous to some users. Related to WCAG 2.0 guideline 2.3. Values should be drawn from the approved vocabulary .
accessibilitySummary	Text	A human-readable summary of specific accessibility features or deficiencies, consistent with the other accessibility metadata but expressing subtleties such as "short descriptions are present but long descriptions will be needed" for non-visual users or "short descriptions are present and no long descriptions are needed".
accountablePerson	Person	Specifies the Person that is legally accountable for the CreativeWork.
acquireLicensePage	CreativeWork or URL	Indicates a page documenting how licenses can be purchased or otherwise acquired, for the current item.
aggregateRating	AggregateRating	The overall rating, based on a collection of reviews or ratings, of the item.

Property	Expected Type	Description
alternativeHeadline	Text	A secondary title of the CreativeWork.
archivedAt	URL or WebPage	Indicates a page or other link involved in archival of a CreativeWork case of MediaReview , the items in a MediaReviewItem may often be inaccessible, but be archived by archival, journalistic, activist, or law enforcement organizations. In such cases, the referenced page may directly publish the content.
assesses	DefinedTerm or Text	The item being described is intended to assess the competency or learning outcome defined by the referenced term.
associatedMedia	MediaObject	A media object that encodes this CreativeWork. This property is a synonym for encoding.
audience	Audience	An intended audience, i.e. a group for whom something was created. Supersedes serviceAudience .
audio	AudioObject or Clip or MusicRecording	An embedded audio object.
author	Organization or Person	The author of this content or rating. Please note that author is specific to this content. HTML5 provides a special mechanism for indicating authorship via the author property. That is equivalent to this and may be used interchangeably.
award	Text	An award won by or for this item. Supersedes awards .
character	Person	Fictional person connected with a creative work.

Property	Expected Type	Description
citation	CreativeWork or Text	A citation or reference to another creative work, such as another published web page, scholarly article, etc.
comment	Comment	Comments, typically from users.
commentCount	Integer	The number of comments this CreativeWork (e.g. Article, Question or Answer) has received. This is most applicable to works published in Web sites with a commenting system; additional comments may exist elsewhere.
conditionsOfAccess	Text	Conditions that affect the availability of, or method(s) of access to, the CreativeWork. Typically used for real world items such as an ArchiveComponent held by an ArchiveOrganization. This property is not suitable for use as a general access control mechanism. It is expressed only in natural language. For example "Available by appointment from the Reading Room" or "Available only from logged-in accounts".
contentLocation	Place	The location depicted or described in the content. For example, the location of a photograph or painting.
contentRating	Rating or Text	Official rating of a piece of content—for example, 'MPAA PG-13'.
contentReferenceTime	DateTime	The specific time described by a creative work, for works (e.g. article objects etc.) that emphasise a particular moment within an Event.
contributor	Organization or Person	A secondary contributor to the CreativeWork or Event.

Property	Expected Type	Description
copyrightHolder	Organization or Person	The party holding the legal copyright to the CreativeWork.
copyrightNotice	Text	Text of a notice appropriate for describing the copyright aspects of the Creative Work, ideally indicating the owner of the copyright for the Work.
copyrightYear	Number	The year during which the claimed copyright for the CreativeWork was asserted.
correction	CorrectionComment or Text or URL	Indicates a correction to a CreativeWork , either via a CorrectionComment textually or in another document.
countryOfOrigin	Country	The country of origin of something, including products as well as creative works such as movie and TV content.
		In the case of TV and movie, this would be the country of the principal production company or individual responsible for the movie. For CreativeWork it is difficult to provide fully general guidance, and such as contentLocation and locationCreated may be more appropriate.
		In the case of products, the country of origin of the product. The exact interpretation of this may vary by context and product type, and can be enumerated here.
creativeWorkStatus	DefinedTerm or Text	The status of a creative work in terms of its stage in a lifecycle. Examples include Incomplete, Draft, Published, Obsolete. Some organizations have their own set of terms for the stages of their publication lifecycle.

Property	Expected Type	Description
creator	Organization or Person	The creator/author of this CreativeWork. This is the same as the Author property for CreativeWork.
creditText	Text	Text that can be used to credit person(s) and/or organization(s) associated with a published Creative Work.
dateCreated	Date or DateTime	The date on which the CreativeWork was created or the item was added to the DataFeed.
dateModified	Date or DateTime	The date on which the CreativeWork was most recently modified or the item's entry was modified within a DataFeed.
datePublished	Date or DateTime	Date of first publication or broadcast. For example the date a CreativeWork was broadcast or a Certification was issued.
digitalSourceType	IPTCDigitalSourceEnumeration	Indicates an IPTCDigitalSourceEnumeration code indicating the nature of the digital source(s) for some CreativeWork .
discussionUrl	URL	A link to the page containing the comments of the CreativeWork.

Property	Expected Type	Description
editEIDR	Text or URL	An EIDR (Entertainment Identifier Registry) identifier representing a specific edit / edition for a work of film or television. For example, the motion picture known as "Ghostbusters" whose titleEIDR is "10.5240/7EC7-228A-510A-053E-CBB8-J" has several edits, e.g. "10.5240/51C5-680A-14C6-E76B-I" and "10.5240/8A35-3BEE-6497-5D12-9E4F". Since schema.org types like Movie and TVEpisode can be used for both a CreativeWork and their multiple expressions, it is possible to use titleEIDR alone (for a general description), or alongside editEIDR for a more edit-specific description.
	Person	Specifies the Person who edited the CreativeWork.
	AlignmentObject	An alignment to an established educational framework.
educationalAlignment		This property should not be used where the nature of the alignment is better described using a simple property, for example to express that a resource teaches or assesses a competency.
educationalLevel	DefinedTerm or Text or URL	The level in terms of progression through an educational or training program. Examples of educational levels include 'beginner', 'intermediate' or 'advanced', and formal sets of level indicators.
educationalUse	DefinedTerm or Text	The purpose of a work in the context of education; for example, 'assigned reading' or 'group work'.
encoding	MediaObject	A media object that encodes this CreativeWork. This property is a synonym for associatedMedia. Supersedes encodings. Inverse property: encodesCreativeWork

Property	Expected Type	Description
encodingFormat	Text or URL	Media type typically expressed using a MIME format (see IANA site a reference), e.g. application/zip for a SoftwareApplication binary, auc for .mp3 etc.
		In cases where a CreativeWork has several media type representa encoding can be used to indicate each MediaObject alongside par encodingFormat information.
		Unregistered or niche encoding and file formats can be indicated ins the most appropriate URL, e.g. defining Web page or a Wikipedia/Wil entry. Supersedes fileFormat .
exampleOfWork	CreativeWork	A creative work that this work is an example/instance/realization/de Inverse property: workExample
expires	Date or DateTime	Date the content expires and is no longer useful or available. For exa VideoObject or NewsArticle whose availability or relevance is tim a ClaimReview fact check whose publisher wants to indicate that it longer be relevant (or helpful to highlight) after some date, or a Certification the validity has expired.
funder	Organization or Person	A person or organization that supports (sponsors) something throu kind of financial contribution.
funding	Grant	A Grant that directly or indirectly provide funding or sponsorship for See also ownershipFundingInfo . Inverse property: fundedItem
genre	Text or URL	Genre of the creative work, broadcast channel or group.

Property	Expected Type	Description
hasPart	CreativeWork	Indicates an item or CreativeWork that is part of this item, or CreativeWork (in some sense). Inverse property: isPartOf
headline	Text	Headline of the article.
inLanguage	Language or Text	The language of the content or performance or used in an action. Please use one of the language codes from the IETF BCP 47 standard . See also availableLanguage . Supersedes language .
interactionStatistic	InteractionCounter	The number of interactions for the CreativeWork using the WebSite SoftwareApplication. The most specific child type of InteractionCounter should be used. Supersedes interactionCount .
interactivityType	Text	The predominant mode of learning supported by the learning resource. Acceptable values are 'active', 'expositive', or 'mixed'.
interpretedAsClaim	Claim	Used to indicate a specific claim contained, implied, translated or rephrased in the content of a MediaObject or other CreativeWork . The interpretation can be indicated using claimInterpreter .
isAccessibleForFree	Boolean	A flag to signal that the item, event, or place is accessible for free. See free .
isBasedOn	CreativeWork or Product or URL	A resource from which this work is derived or from which it is a modification or adaptation. Supersedes isBasedOnUrl .
isFamilyFriendly	Boolean	Indicates whether this content is family friendly.

Property	Expected Type	Description
isPartOf	CreativeWork or URL	Indicates an item or CreativeWork that this item, or CreativeWork (in sense), is part of. Inverse property: hasPart
keywords	DefinedTerm or Text or URL	Keywords or tags used to describe some item. Multiple textual entries in a keywords list are typically delimited by commas, or by repeating the
learningResourceType	DefinedTerm or Text	The predominant type or kind characterizing the learning resource. For example, 'presentation', 'handout'.
license	CreativeWork or URL	A license document that applies to this content, typically indicated by
locationCreated	Place	The location where the CreativeWork was created, which may not be the same as the location depicted in the CreativeWork.
mainEntity	Thing	Indicates the primary entity described in some page or other CreativeWork. Inverse property: mainEntityOfPage
maintainer	Organization or Person	A maintainer of a Dataset , software package (SoftwareApplication), or other Project . A maintainer is a Person or Organization that makes contributions to, and/or publication of, some (typically complex) artifact. It is common for distributions of software and data to be based on "upstream" sources. When maintainer is applied to a specific version of some artifact, a particular version or packaging of a Dataset , it is always possible that the upstream source has a different maintainer. The isBasedOn property is used to indicate such relationships between datasets to make the dependencies and maintenance roles clear. Similarly in the case of software, a package

Property	Expected Type	Description
		dedicated maintainers working on integration into software distributions as Ubuntu, as well as upstream maintainers of the underlying work.
material	Product or Text or URL	A material that something is made from, e.g. leather, wool, cotton, paper.
materialExtent	QuantitativeValue or Text	The quantity of the materials being described or an expression of the space they occupy.
mentions	Thing	Indicates that the CreativeWork contains a reference to, but is not necessarily about a concept.
offers	Demand or Offer	An offer to provide this item—for example, an offer to sell a product, rent a DVD of a movie, perform a service, or give away tickets to an event. Use businessFunction to indicate the kind of transaction offered, i.e. sell , lease , etc. This property can also be used to describe a Demand . While this property is listed as expected on a number of common types, it can be used in cases where that case, using a second type, such as Product or a subtype of Product , to clarify the nature of the offer. Inverse property: itemOffered
pattern	DefinedTerm or Text	A pattern that something has, for example 'polka dot', 'striped', 'Canadian flag'. Values are typically expressed as text, although links to controlled vocabularies and schemes are also supported.
position	Integer or Text	The position of an item in a series or sequence of items.

Property	Expected Type	Description
producer	Organization or Person	The person or organization who produced the work (e.g. music album producer, TV/radio series etc.).
provider	Organization or Person	The service provider, service operator, or service performer; the goods or services producer. Another party (a seller) may offer those services or goods on behalf of the provider. A provider may also serve as the seller. Supersedes creator .
publication	PublicationEvent	A publication event associated with the item.
publisher	Organization or Person	The publisher of the creative work.
publisherImprint	Organization	The publishing division which published the comic.
publishingPrinciples	CreativeWork or URL	The publishingPrinciples property indicates (typically via URL) a document describing the editorial principles of an Organization (or individual Person writing a blog) that relate to their activities as a publisher, e.g. editorial, diversity policies. When applied to a CreativeWork (e.g. NewsArticle) the principles are those of the party primarily responsible for the creation of the CreativeWork. While such policies are most typically expressed in natural language, sometimes related information (e.g. indicating a funder) can be expressed using schema.org terminology.
recordedAt	Event	The Event where the CreativeWork was recorded. The CreativeWork may capture all or part of the event. Inverse property: recordedIn
releasedEvent	PublicationEvent	The place and time the release was issued, expressed as a PublicationEvent.

Property	Expected Type	Description
review	Review	A review of the item. Supersedes reviews .
schemaVersion	Text or URL	Indicates (by URL or string) a particular version of a schema used in a CreativeWork. This property was created primarily to indicate the use of a specific schema.org release, e.g. 10.0 as a simple string, or more explicitly a URL, https://schema.org/docs/releases.html#v10.0 . There are many situations in which other schemas might usefully be referenced this way. See http://dublincore.org/specifications/dublin-core/dceschema/07-02/ but this has not been carefully explored in the community.
sdDatePublished	Date	Indicates the date on which the current structured data was generated and published. Typically used alongside sdPublisher .
sdLicense	CreativeWork or URL	A license document that applies to this structured data, typically in the form of a URL.
sdPublisher	Organization or Person	Indicates the party responsible for generating and publishing the current structured data markup, typically in cases where the structured data is generated automatically from existing published content but published on a different platform. For example, student projects and open data initiatives often re-publish existing content with more explicitly structured metadata. The sdPublisher property helps make such practices more explicit.
size	DefinedTerm or QuantitativeValue or SizeSpecification or Text	A standardized size of a product or creative work, specified either through a simple textual string (for example 'XL', '32Wx34L'), a QuantitativeValue with a unitCode, or a comprehensive and structured SizeSpecification . In some cases, the width , height , depth and weight properties may be more applicable.
sourceOrganization	Organization	The Organization on whose behalf the creator was working.

Property	Expected Type	Description
spatial	Place	The "spatial" property can be used in cases when more specific properties (locationCreated , spatialCoverage , contentLocation) are not known to be appropriate.
spatialCoverage	Place	The spatialCoverage of a CreativeWork indicates the place(s) which is/are the focus of the content. It is a subproperty of contentLocation intended for more technical and detailed materials. For example with a Dataset it indicates areas that the dataset describes: a dataset of New York would have spatialCoverage which was the place: the state of New York.
sponsor	Organization or Person	A person or organization that supports a thing through a pledge, promise, or financial contribution. E.g. a sponsor of a Medical Study or a corporation sponsoring an event.
teaches	DefinedTerm or Text	The item being described is intended to help a person learn the complex learning outcome defined by the referenced term.
temporal	DateTime or Text	The "temporal" property can be used in cases where more specific properties (e.g. temporalCoverage , dateCreated , dateModified , datePublished) are not known to be appropriate.
temporalCoverage	DateTime or Text or URL	The temporalCoverage of a CreativeWork indicates the period that the work applies to, i.e. that it describes, either as a DateTime or as a textual string indicating a time period in ISO 8601 time interval format . In the case of a Dataset it will typically indicate the relevant time period in a precise manner (e.g. for a 2011 census dataset, the year 2011 would be written "2011/12/31"). Other forms of content, e.g. ScholarlyArticle, Book, TVSeries or TVEpisode, indicate their temporalCoverage in broader terms - textually or via a URL. Written works such as books may sometimes have precise temporal coverage too, e.g. a work set in 1939 - 1945 can be indicated in ISO 8601

Property	Expected Type	Description
		that can be commercially licensed, <code>usageInfo</code> can provide detailed, more specific information about licensing options. This property can be used alongside the <code>license</code> property which indicates the license(s) applicable to some piece of content. The <code>usageInfo</code> properties provide information about other licensing options, e.g. acquiring commercial usage rights for an image that is also available under non-commercial commons licenses.
version	Number or Text	The version of the CreativeWork embodied by a specified resource.
video	Clip or VideoObject	An embedded video object.
workExample	CreativeWork	Example/instance/realization/derivation of the concept of this creative work. E.g. the paperback edition, first edition, or e-book. Inverse property: exampleOfWork
workTranslation	CreativeWork	A work that is a translation of the content of this work. E.g. an English workTranslation “Journey to the West”, a German workTranslation “Monkeys Pilgerfahrt” and a Vietnamese translation Tây du ký bình khảo. Inverse property: translationOfWork
Properties from Thing		
additionalType	Text or URL	An additional type for the item, typically used for adding more specific types from external vocabularies in microdata syntax. This is a relationship between something and a class that the thing is in. Typically the value is a URI or RDF class, and in this case corresponds to the use of <code>rdf:type</code> in RDF. Values can be used sparingly, for cases where useful information can

Property	Expected Type	Description
		without their being an appropriate schema to reference. In the case of <code>ImageObject</code> values, the class label should follow the schema.org style guide .
alternateName	Text	An alias for the item.
description	Text or TextObject	A description of the item.
disambiguatingDescription	Text	A sub property of description. A short description of the item used to disambiguate from other, similar items. Information from other properties (e.g. particular, name) may be necessary for the description to be useful for disambiguation.
identifier	PropertyValue or Text or URL	The identifier property represents any kind of identifier for any kind of entity, such as ISBNs, GTIN codes, UUIDs etc. Schema.org provides dedicated properties for representing many of these, either as textual strings or as URL (URI) links. See background notes for more details.
image	ImageObject or URL	An image of the item. This can be a URL or a fully described ImageObject .
mainEntityOfPage	CreativeWork or URL	Indicates a page (or other CreativeWork) for which this thing is the main entity being described. See background notes for details. Inverse property: mainEntity
name	Text	The name of the item.
potentialAction	Action	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.

Property	Expected Type	Description
sameAs	URL	URL of a reference Web page that unambiguously indicates the item E.g. the URL of the item's Wikipedia page, Wikidata entry, or official website
subjectOf	CreativeWork or Event	A CreativeWork or Event about this Thing. Inverse property: about
url	URL	URL of the item.

Source

<https://github.com/schemaorg/schemaorg/issues/2140>